

Database Management System for E-Commerce Platform

<i>Priyanshu Jha</i>	<i>Dhruv Marwal</i>	<i>Shivang Jain</i>
<i>BTech Artificial Intelligence</i>	<i>BTech Artificial Intelligence</i>	<i>BTech Artificial Intelligence</i>
<i>Mukesh Patel School of</i>	<i>Mukesh Patel School of</i>	<i>Mukesh Patel School of</i>
<i>Technology Management and</i>	<i>Technology Management and</i>	<i>Technology Management and</i>
<i>Engineering</i>	<i>Engineering</i>	<i>Engineering</i>
<i>Mumbai, India</i>	<i>Mumbai, India</i>	<i>Mumbai, India</i>
<i>priyanshu.jha17@nmims.in</i>	<i>dhruv.marwal59@nmims.in</i>	<i>shivang.jain60@nmims.in</i>

Abstract

The project proposes a relational database model for an e-commerce platform. The primary goal is to streamline product listings, user interactions, orders, and shipment processes. The developed Entity-Relationship Diagram (ERD) defines multiple interconnected entities, ensuring effective data handling, minimizing redundancy, and maximizing integrity.

Index Terms

E-commerce, ERD, DBMS, relational database, database design.

I. INTRODUCTION

E-commerce platforms rely on comprehensive database systems to manage vast data volumes efficiently. This paper presents a robust relational model designed using an ERD for an online shopping platform. The system includes users, products, categories, brands, suppliers, orders, carts, payments, warehouses, and shipment processes, creating a cohesive digital commerce environment.

II. METHODOLOGY

The ERD was created following standard relational database principles. Primary keys, foreign keys, and entity relationships were carefully designed to ensure minimal redundancy, maintain normalization, and improve scalability.

III. SYSTEM DESIGN

A. Entities and Attributes

- User: Attributes include UserID, Name, Username, Age, Phone, Email, Address (composed of HouseNo, Building, Street, Locality).
- Product: Defined by ProductID, Name, Description, Specifications, Productivity, and Images.
- Brand: Characterized by BrandID and BrandName.
- Category: Identified by CategoryID and CategoryName.
- Supplier: SupplierID, Name, Phone, Email, Address.
- Shipment: ShipmentID, Location, Estimated Delivery, and associated Shipping Company.
- Warehouse: WarehouseID, Location, and Capacity.
- Order: OrderID, UserID, ProductID, ShipmentID, and PlatformID.
- Cart: UserID, ProductID.
- Seller: SellerID, Name, Phone, Email, and Address.
- Payment: PaymentID, OrderID, Amount, Method, Date.
- Platform: PlatformID, Name, and URL.

B. Relationships

- A User places multiple Orders.
- A User maintains a Cart with multiple Products.
- An Order includes one or more Products and is linked to Shipments.
- Products are supplied by Suppliers and stored in Warehouses.
- Products belong to a Brand and are categorized under Categories.
- Shipments are handled by specific Shipping Companies.
- Payments are associated with Orders.
- Sellers set Products on various Platforms.

IV. Questions and Queries

Q1. Product details for those product which has maximum order and the details of supplier and seller.

```
select pi.name as ProductName,pi.Description,pi.Specifications,count(o.OrderID) as
totalorders,s.name as suppliername,s.phone as supplierphone,
sel.name as sellername,sel.phone as sellerphone
from `order` o
join product p on o.productid=p.productid
join productinfo pi on p.productinfoid=pi.productinfoid
```

```

join product_supplier ps on p.productid = ps.productID
join supplier s on ps.supplierid = s.supplierid
join seller_product sp on p.productid = sp.productid
join seller sel on sp.sellerid = sel.sellerid
where o.productid in (select productid from `order` group by productid having
count(orderid)=(select max(order_count) from (select count(orderid) as order_count from
`order` group by productid) as sub)) group by pi.name, s.name,sel.name, pi.description,
pi.specifications, s.phone, sel.phone;

```

Output:

	ProductName	Description	Specifications	totalorders	suppliername	supplierphone	sellername	sellerphone
▶	Samsung Galaxy S21	Android smartphone	8GB RAM, 128GB Storage	1	TechSupply	8888811111	QuickSell	7777711111
	iPhone 14	iOS smartphone	128GB, A15 Bionic	1	TechSupply	8888811111	TopDealz	7777722222
	Dell Inspiron	Dell laptop	i7, 512GB SSD	1	TechSupply	8888811111	MegaStore	7777733333
	OnePlus 10	Android smartphone	12GB RAM, 256GB Storage	1	GadgetWorld	8888822222	QuickSell	7777711111
	MacBook Air	Apple laptop	M2 Chip, 512GB SSD	1	GadgetWorld	8888822222	TopDealz	7777722222
	iPad Air	Apple tablet	64GB, WiFi	1	GadgetWorld	8888822222	MegaStore	7777733333
	HP Pavilion	HP laptop	i5, 16GB RAM, 1TB SSD	1	DigitalHub	8888833333	QuickSell	7777711111
	Sony WH-1000XM4	Headphones	Bluetooth, Noise Cancelling	1	DigitalHub	8888833333	MegaStore	7777733333

Q2. Product info details where shipment date is between 15 and 18.

```

select pi.productinfoid, pi.name as productname, pi.description, pi.specifications,
pi.images,s.shipmentid as sid, s.estimateddelivery as shipmentdate
from shipment s
join `order` o on s.shipmentid = o.shipmentid
join product p on o.productid= p.productid
join productinfo pi on p.productinfoid = pi.productinfoid
where s.estimateddelivery between '2025-04-15' and '2025-04-18';

```

Output:

	productinfoid	productname	description	specifications	images	sid	shipmentdate
▶	1	Samsung Galaxy S21	Android smartphone	8GB RAM, 128GB Storage	img_s21.jpg	1	2025-04-15
	2	iPhone 14	iOS smartphone	128GB, A15 Bionic	img_iphone14.jpg	2	2025-04-17
	3	OnePlus 10	Android smartphone	12GB RAM, 256GB Storage	img_op10.jpg	3	2025-04-18
	4	MacBook Air	Apple laptop	M2 Chip, 512GB SSD	img_mac.jpg	2	2025-04-17
	6	Sony WH-1000XM4	Headphones	Bluetooth, Noise Cancelling	img_sony.jpg	1	2025-04-15
	7	Dell Inspiron	Dell laptop	i7, 512GB SSD	img_dell.jpg	2	2025-04-17
	8	iPad Air	Apple tablet	64GB, WiFi	img_ipad.jpg	3	2025-04-18

#Sample Queries:

Q3. TOP 3 Expensive order.

```
select o.orderid, u.name as username, pi.name as productname, p.amount
from `order` o
join user u on o.userid = u.userid
join product pr on o.productid = pr.productid
join productinfo pi on pr.productinfoid = pi.productinfoid
join payment p on o.orderid = p.orderid
order by p.amount desc limit 3;
```

	OrderID	UserName	ProductName	Amount
▶	4	Diana Patel	MacBook Air	119999.00
	2	Bob Smith	iPhone 14	79999.00
	1	Alice Johnson	Samsung Galaxy S21	69999.00

Q4. Sellers selling products in 'Laptops' category.

```
select distinct s.name as sellername, pi.name as productname, c.categoryname
from seller s
join seller_product sp on s.sellerid = sp.sellerid
join product p on sp.productid = p.productid
join category c on p.categoryid = c.categoryid
join productinfo pi on p.productinfoid = pi.productinfoid
where c.categoryname = 'laptops';
```

	SellerName	ProductName	CategoryName
▶	TopDealz	MacBook Air	Laptops
	QuickSell	HP Pavilion	Laptops
	MegaStore	Dell Inspiron	Laptops

Q5. Total revenue generated by each platform.]

```
select pl.name as platformname, sum(pay.amount) as totalrevenue
from `order` o
join platform pl on o.platformid = pl.platformid
join payment pay on o.orderid = pay.orderid
group by pl.platformid;
```

	PlatformName	TotalRevenue
▶	Amazon	129997.00
	Flipkart	239997.00
	Snapdeal	95998.00

Q6. Most active users (by number of orders).

```
select u.name as username, count(o.orderid) as totalorders
from user u
join `order` o on u.userid = o.userid
group by u.userid
order by totalorders desc;
```

	UserName	TotalOrders
▶	Alice Johnson	1
	Bob Smith	1
	Charlie Davis	1
	Diana Patel	1
	Ethan Brown	1
	Fiona Lee	1
	George Kim	1
	Hannah Scott	1

Q7. Average product price by category.'

```
select c.categoryname, round(avg(pay.amount), 2) as avgprice
from `order` o
join payment pay on o.orderid = pay.orderid
join product p on o.productid = p.productid
join category c on p.categoryid = c.categoryid
group by c.categoryid;
```

	CategoryName	AvgPrice
▶	Smartphones	68332.33
	Laptops	76665.67
	Headphones	4999.00
	Tablets	25999.00

Q8. Suppliers supplying to most number of unique products.

```
select s.name as suppliername, count(distinct ps.productid) as productssupplied
from supplier s
join product_supplier ps on s.supplierid = ps.supplierid
group by s.supplierid
order by productssupplied desc;
```

	SupplierName	ProductsSupplied
►	TechSupply	3
	GadgetWorld	3
	DigitalHub	2

Q9. Users who purchased and reviewed the same product.

```
select u.name as username, pi.name as productname
from user u
join `order` o on u.userid = o.userid
join review r on u.userid = r.userid and o.productid = r.productid
join product p on r.productid = p.productid
join productinfo pi on p.productinfoid = pi.productinfoid;
```

	UserName	ProductName
►	Alice Johnson	Samsung Galaxy S21
	Bob Smith	iPhone 14
	Charlie Davis	OnePlus 10
	Diana Patel	MacBook Air
	Ethan Brown	HP Pavilion
	Fiona Lee	Sony WH-1000XM4
	George Kim	Dell Inspiron
	Hannah Scott	iPad Air

IV. RESULTS

The ERD successfully models the business logic of an e-commerce platform, covering inventory management, order processing, shipment tracking, and payment handling. The relationships are normalized to third normal form (3NF) to reduce data redundancy and improve query performance.

V. CONCLUSION

The proposed ERD structure provides a scalable and efficient framework for e-commerce databases. It supports the dynamic requirements of an online shopping system, ensuring data integrity and system flexibility.

VI. FUTURE SCOPE

- Integration with AI for personalized recommendations.
- Support for blockchain-based secure transactions.
- Scalability for multi-vendor and international marketplace operations.
- Incorporation of real-time analytics for user behavior.

VII. LIMITATIONS

- The current design does not include detailed inventory replenishment mechanisms.
- Real-time warehouse stock syncing across multiple platforms requires additional optimization.